

Fermion-Boson Two-Body Model of Quarks and Leptons and Cabibbo Mixing.

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Summary. — Quark and lepton masses and the Cabibbo angle are estimated on the basis of a subquark model where leptons and quarks are made of a colour- $\underline{3}^*$ fermion F and a colour- $\underline{3}$ boson H as (FH) and $(F\bar{H})$, respectively.

Recently a new mechanism for Cabibbo mixing has been proposed on the basis of a composite model of quarks and leptons ⁽¹⁾. In the model, there are three subquarks (so-called «hakams» ⁽²⁾) h_i ($i = 1, 2, 3$) which carry «generation» quantum number, and a lepton contains two hakams (hh), so that 3-generations of leptons are understood as $\underline{3}^*$ in $\underline{3} \times \underline{3} = \underline{3}^* + \underline{6}$ of SU_3 -generation symmetry, while a quark contains ($h\bar{h}$), so that 3-generations of quarks are explained as the three diagonal elements

$$\pi^8 = (h_1\bar{h}^1 - h_2\bar{h}^2)/\sqrt{2}, \quad \eta^8 = (h_1\bar{h}^1 + h_2\bar{h}^2 - 2h_3\bar{h}^3)/\sqrt{6} \quad \text{and} \quad \eta^1 = (h_1\bar{h}^1 + h_2\bar{h}^2 + h_3\bar{h}^3)/\sqrt{3}.$$

The existence of a $SU_{3,G}$ -symmetry breaking induces, for leptons, merely mass difference among those, while, for quarks, it causes π^8 - η^8 mixing in addition to the mass differences. The model predicts the relation

$$(1) \quad \tan \theta_c = \sqrt{3} \frac{\sqrt{m_\mu} - \sqrt{m_e}}{2\sqrt{m_\tau} - \sqrt{m_\mu} - \sqrt{m_e}} = 0.225,$$

which is in excellent agreement with experiment.

In the model, however, eq. (1) has been essentially derived from the π^8 - η^8 mixing in the down-quarks, and that the π^8 - η^8 mixing in the up-quarks is negligibly small is nothing but an assumption. The model provides no predictions for up-lepton (neutrino) masses and up-quark masses, and even for down-quark masses. In the model, a fermion (so-called «wakem» ⁽²⁾), which carries weak quantum number, inside the lepton is different from that inside the quark. The model provides many exotic quarks

⁽¹⁾ Y. KOIDE: *Phys. Rev. Lett.*, **47**, 1241 (1981)

⁽²⁾ H. TERAZAWA: *Phys. Rev. D*, **22**, 184 (1980).

and leptons which are unwelcome and, therefore, to suppress those states, we have to introduce many other hypothetical forces besides the hypothetical force which selects three $i = j$ pairs out of the nonet states $(h_i \bar{h}^j)$.

In the present paper, we inherit the idea that quarks (u, d), (c, s) and (t, b) are assigned to the diagonal elements π^8 , η^8 and η^1 of $SU_{3,G} \times 3^*$, respectively, but we propose another composite model where quarks and leptons are two-body bound systems of a fermion and a boson, and we give the prediction of quark and lepton masses in addition to the derivation of eq. (1).

In our model, there are a colour- 3^* fermion $F_i^a = (U_i^a, D_i^a)$ and a colour-3 boson H_{ai} , where a and i are SU_3 -colour and SU_3 -generation (*) indices, respectively. The left-handed components of F_i^a form doublets of gauge group $SU_{2\text{weak}}$. The electric charges of F and H are $(\frac{1}{3}, -\frac{2}{3})$ and $-\frac{1}{3}$, respectively (**). The lepton ℓ_i and the quark q_i are explained as colour-singlet state

$$(F_i H_i) = \sum_a F_i^a H_{ai} / \sqrt{3}$$

and colour-triplet state

$$(F_i \bar{H}^i)_a = \sum_{b,c} \varepsilon_{abc} F_i^b \bar{H}^{ci} / \sqrt{6},$$

respectively.

To select the three $i = j$ pairs out of $(F_i \bar{H}^j)$, we introduce a superstrong confinement potential V_s which is provided by interactions (***) with scalar bosons φ_s ($i = 1, 2, 3$):

$$(2) \quad H_s = g_s \sum_i (\bar{F}^i F_i - H^{*i} H_i) \varphi_s^i.$$

Interaction (2) provides the potential $-\delta_{ij} V_s$ for the systems $(F_i H_j)$ and $(F_i \bar{H}^j)$, while it provides $+\delta_{ij} V_s$ for the systems $(F_i F_j)$, $(F_i \bar{F}^j)$, $(H_i H_j)$ and $(H_i \bar{H}^j)$. Since we assume the confinement due to the potential $-V_s$, the allowed states are only $(F_i H_j)$ and $(F_i \bar{H}^j)$ with $i = j$. The colour-potentials in the colour-singlet state $(F_i H_i)$ and the colour-triplet state $(F_i \bar{H}^i)_a$ are negative, that is, $-4V_c/3$ and $-2V_c/3$, respectively, while those in the colour-8 state $(F_i H_i)_a^b$ and the colour- 6^* state $(F_i \bar{H}^i)^{(ab)}$ are positive, that is, $V_c/6$ and $V_c/3$, respectively. Thus we consider that physical states are only the three colour-singlet states $(F_i H_i)$ ($i = 1, 2, 3$) and the three colour-triplet states $(F_i \bar{H}^i)_a$ ($i = 1, 2, 3$). (Note that, in the previous model⁽¹⁾, leptons belong to $SU_{3,G} \times 3^*$, while, in the present model, leptons belong to $SU_{3,G}$.)

Furthermore, we assume the existence of $SU_{3,G}$ -invariant interactions which make only $\eta^1 = (q_1 + q_2 + q_3)/\sqrt{3}$ massive and leave $SU_{3,G}$ -6 leptons and SU_3 -8 quarks massless. For example, we introduce the interactions mediated by scalar bosons $\varphi_G^{(1)j}$ and $\varphi_G^{(8)b}$ which belong to $(1, 8)$ and $(8, 1)$ of (SU_{3c}, SU_{3G}) :

$$(3) \quad H_{G\text{-inv}} = g_G \sum_a \sum_{ij} (\bar{F}_i^a F_j^a - H^{*ai} H_{aj}) \varphi_G^{(1)j} + \frac{1}{2} g_G \sum_i \sum_{ab} (\bar{F}_i^a F_i^b - H^{*bi} H_{ai}) \varphi_G^{(8)a}.$$

(*) This symmetry $SU_{3,G}$ is not a local gauge group, but merely a phenomenological symmetry which is badly broken.

(**) Since the fundamental left-handed fermions show $\sum Q_F \neq 0$, the model is not anomaly-free (*). We must consider the $SU_{2L} \times SU_{2R} \times U_1$ gauge model⁽⁴⁾ with $M_{WR} \gg m_{WL}$ as electroweak interaction.

(3) D. J. GROSS and R. JACKIW: *Phys. Rev. D*, **6**, 477 (1972).

(4) J. C. PATI and A. SALAM: *Phys. Rev. D*, **10**, 275 (1974); R. N. MOHAPATRA and J. C. PATI: *Phys. Rev. D*, **11**, 566 (1975); G. SENJANOVIĆ and R. N. MOHAPATRA: *Phys. Rev. D*, **12**, 1502 (1975).

(***) Of course, interaction (2) is not $SU_{3,G}$ -invariant.

Here the factor $\frac{1}{2}$ in the $(8, \underline{1})$ -interaction has phenomenologically been chosen as $(\underline{1}, \underline{6})$ -leptons and $(\underline{3}, \underline{8})$ -quarks are massless, that is the potential terms in those states are $-\frac{1}{3} + \frac{1}{4} \cdot \frac{4}{3} = 0$ and $-\frac{1}{6} + \frac{1}{4} \cdot \frac{2}{3} = 0$, respectively.

If there are small $SU_{3,G}$ -symmetry-breaking mass terms, then the mass matrices of down-leptons e_i ($e_1 = e$, $e_2 = \mu$, $e_3 = \tau$) and down-quarks d_i are given by

$$(4) \quad M^e = \begin{pmatrix} m_{e_1} & 0 & 0 \\ 0 & m_{e_2} & 0 \\ 0 & 0 & m_{e_3} \end{pmatrix}, \quad M^d = \begin{pmatrix} m_{d_1} & 0 & 0 \\ 0 & m_{d_2} & 0 \\ 0 & 0 & m_{d_3} \end{pmatrix} + A^d \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix},$$

(those of up-leptons and up-quarks are given similarly to (4).) If we build the model where

$$(5) \quad m_{e_i} \propto (z_i + z_0)^2, \quad m_{d_i} \propto z_i^2,$$

$$(6) \quad z_1 + z_2 + z_3 = 0,$$

and z_0 is a constant, we can get the relation for the π^8 - η^8 mixing angle θ^d in down-quarks

$$(7) \quad \tan 2\theta^d = -\sqrt{3} \frac{m_{d_1} - m_{d_2}}{m_{d_1} + m_{d_2} - 2m_{d_3}} = -\sqrt{3} \frac{z_1^2 - z_2^2}{z_1^2 + z_2^2 - 2z_3^2},$$

or

$$(8) \quad \tan \theta^d = \sqrt{3} \frac{z_2 - z_1}{2z_3 - z_2 - z_1} = \sqrt{3} \frac{\sqrt{m_\mu} - \sqrt{m_e}}{2\sqrt{m_\tau} - \sqrt{m_\mu} - \sqrt{m_e}}.$$

Therefore, if, in the model, we can show a negligibly small u-c mixing angle $\theta^u \simeq 0$ together with a large mass difference $m_c - m_u$ and $m_{\nu_i} \simeq 0$, outlines of our scenarios will be accomplished.

We assume that the $SU_{3,G}$ -symmetry-breaking mass terms m_{l_i} and m_{q_i} in eq. (4) are dynamically generated by an interaction which is mediated by a vector boson φ_F . Hereafter we call the coupling constants of the fundamental constituents F_i and H_i with the boson φ_F «flavour charges» Q_F . We assume that the mass m_{l_i} (m_{q_i}) of the composite (FH) ((F $\bar{H}$)) is given by the sum of the potential term $+Q_F(F)Q_F(H)v_F$ ($-Q_F(F)Q_F(H)v_F$) and the self-energy terms $\frac{1}{2}[Q_F(F)]^2v_F + \frac{1}{2}[Q_F(H)]^2v_F$, that is by

$$(9) \quad m \begin{pmatrix} FH \\ F\bar{H} \end{pmatrix} = \frac{1}{2}[Q_F(F) \pm Q_F(H)]^2v_F,$$

where v_F is a parameter which depends on the «effective radius» r_0 of the system (for example, such a parameter as $v_F(r_0) = 1/4\pi r_0$ if the potential is Coulomb-like). Here we have assumed that the self-energy terms of the constituents are characterized by the same parameter $v_F(r_0)$ as in the potential between the constituents.

The charge assignment which leads to $m_{\nu_i} = 0$ and relation (5) is the following:

$$(10) \quad \begin{cases} Q_F(U_i) = -Q_F(H_i) = \alpha z_i + \frac{1}{2}(z_i + z_0), \\ Q_F(D_i) = \alpha z_i - \frac{1}{2}(z_i + z_0), \end{cases}$$

where α is a constant. Then we get

$$(11) \quad \begin{cases} m_{\nu_i} = 0, & m_{u_i} = [(2\alpha + 1)z_i + z_0]^2 v_F/2, \\ m_{e_i} = (z_i + z_0)^2 v_F/2, & m_{d_i} = (2\alpha z_i)^2 v_F/2, \end{cases}$$

and, therefore, we can derive relation (8). Phenomenologically, we take a great interest in the case (*) $\alpha = \frac{1}{3}$ which predicts

$$(12) \quad \theta^u = 0.011,$$

$$(13) \quad \begin{cases} m_u = 0.070 \text{ GeV}, & m_c = 2.317 \text{ GeV}, \\ m_d = 0.070 \text{ GeV}, & m_s = 0.210 \text{ GeV}, \end{cases}$$

under the approximation $A^u \gg m_{u_i}$ and $A^d \gg m_{d_i}$, although the predicted value of m_c is somewhat large. It is likely that the effective radius of the lepton is small relatively to that of the quark, that is $r_0^l < r_0^q$, and therefore that $v_F(r_0^l) > v_F(r_0^q)$. If we take $v_F(r_0^q)/v_F(r_0^l) \simeq \frac{2}{3}$, we get

$$(14) \quad \begin{cases} m_u = 0.047 \text{ GeV}, & m_c = 1.545 \text{ GeV}, \\ m_d = 0.047 \text{ GeV}, & m_s = 0.140 \text{ GeV}, \end{cases}$$

instead of prediction (13). (Of course, there is no theoretical ground for the choice $\alpha = \frac{1}{3}$ and $v_F^q/v_F^l = \frac{2}{3}$. We can merely claim that there is a favourable solution of the parameters α and v_F^q/v_F^l .) The predicted value (12) of θ^u is desirably small and, moreover, consideration of the mixing with the $SU_{3,c}$ -singlet state (t, b) shifts the predicted value (8) of θ^d by $\Delta\theta^d \simeq 0.01$ (input $m_b \simeq 4.5 \text{ GeV}$) so that $\theta_c = \theta^d - \theta^u$ is again given by eq. (1).

Finally, we propose another model, where $v_F^q = v_F^l$. In the previous model, the choice of $v_F^q/v_F^l \simeq \frac{2}{3}$ seems to be artificial. We introduce a $SU_{3,c}$ -octet boson $\varphi_F^{(8)}$ in addition to the previous $SU_{3,c}$ -singlet boson φ_F and we assume the existence of the two kinds of flavour charges Q_F and $Q_F^{(8)}$. Since the self-energy term, the colour-singlet potential term and the colour-triplet potential term take the coefficients $\frac{2}{3}$, $-\frac{4}{3}$ and $-\frac{2}{3}$, respectively, the contribution from $Q_F^{(8)}$ to the colour-singlet (FH) and the colour-triplet (FH) are $\frac{2}{3}[Q_F(F) - Q_F(\bar{H})]^2 v_F$ and $\frac{2}{3}\{[Q_F(F)]^2 + [Q_F(\bar{H})]^2 - Q_F(F)Q_F(\bar{H})\} v_F$, respectively. We assign the flavour charges $Q_F^{(8)}$ as

$$(15) \quad Q_F^{(8)}(U_i) = Q_F^{(8)}(D_i) = Q_F^{(8)}(\bar{H}_i) = \beta z_i.$$

Then we get

$$(16) \quad \begin{cases} m_{\nu_i} = 0, & m_{u_i} = (z_i + z_0)^2 + \frac{4}{3}(\beta z_i)^2, \\ m_{e_i} = (z_i + z_0)^2, & m_{d_i} = \frac{4}{3}(\beta z_i)^2, \end{cases}$$

where we drop the common coefficient $v_F/2$ and we put $\alpha = 0$. We suppose that the

(*) Other cases of α are, for example, as follows: $\theta^u = 0.003$, $m_u = 0.052 \text{ GeV}$, $m_c = 2.002$, $m_d = 0.039 \text{ GeV}$ and $m_s = 0.118 \text{ GeV}$ for $\alpha = \frac{1}{4}$; $\theta^u = -0.006$, $m_u = 0.043 \text{ GeV}$, $m_c = 1.713 \text{ GeV}$, $m_d = 0.018 \text{ GeV}$ and $m_s = 0.053 \text{ GeV}$ for $\alpha = \frac{1}{6}$.

flavour charge Q_F is given by the $\pi/4$ mixing between a $SU_{3,G}$ -octet current

$$(17) \quad (z_1^2 + z_2^2 + z_3^2)^{-\frac{1}{2}} \sum_i z_i \sum_a (\bar{U}_a^i U_i^a - \bar{D}_a^i D_i^a)$$

and a $SU_{3,G}$ -singlet current

$$(18) \quad (1/\sqrt{3}) \sum_i \sum_a (\bar{U}_a^i U_i^a - \bar{D}_a^i D_i^a),$$

while $Q_F^{(8)}$ is given by the coloured $SU_{3,G}$ -octet current

$$(19) \quad (z_1^2 + z_2^2 + z_3^2)^{-\frac{1}{2}} \sum_i z_i (\bar{U}_a^i U_i^b + \bar{D}_a^i D_i^b).$$

If, on the analogy of eq. (3), we assume the existence of the factor $\frac{1}{2}$ for the current (19), we get $\beta = 1/\sqrt{2}$ together with

$$(20) \quad z_0^2 = (z_1^2 + z_2^2 + z_3^2)/3.$$

The choice of $\beta = 1/\sqrt{2}$ predicts

$$(21) \quad \theta^u = 0.011,$$

$$(22) \quad \begin{cases} m_u = 0.168 \text{ GeV}, & m_c = 1.517 \text{ GeV}, \\ m_d = 0.105 \text{ GeV}, & m_s = 0.316 \text{ GeV}, \end{cases}$$

which are plausible as quark masses. Note that relation (20) predicts, from the experimental values of m_e and m_μ ,

$$(23) \quad m_\tau = 1.777 \text{ GeV},$$

which is in good agreement with the observed value ⁽⁵⁾ $m_\tau = 1.784 \pm 0.004 \text{ GeV}$.

In conclusion, we have predicted reasonable values of the Cabibbo angle θ_C and quark and lepton masses by assuming that masses of the composite systems are generated only by $SU_{3,G}$ -symmetry-breaking interactions through «flavour charges». We have proposed two models for the flavour-charge assignments which can predict large mass $m_c \simeq 1.5 \text{ GeV}$ relatively to m_u , m_d and m_s in spite of $m_{\nu_e} = m_{\nu_\mu} = m_{\nu_\tau} = 0$.

⁽⁵⁾ PARTICLE DATA GROUP: *Rev. Mod. Phys.*, **52**, s1 (1980).